

Cycles 1-3 Clone 0 Report — M-DAW-SPIKE-1/palette-schema-v2-hydration-render (Fork 07063458736e)

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Cycles 1-3 Clone 0 Report — M-DAW-SPIKE-1/palette-schema-v2-hydration-render (Fork 07063458736e)

Abstract

Cycles 1-3 of clone-0 (fork 07063458736e) close the cycle-35 Branch A first substantive activation of the c34 palette_v2 schema in an actual render at **RENDER_FAILS** — a first-class negative finding under the pre-registered 3-verdict rubric. Cycle 1 executed the full pipeline with an honest failure diagnosis: VST3-binary-internal nondeterminism (a fair c31 STILL_GAP extension), not a c33 P1 hydration defect. Cycles 2-3 are c30-codified standby-held VALIDATED re-affirmations (three consecutive standby resolutions total; no workspace state changes; no re-emissions; anti-pattern locks intact). The negative finding preserves substantial positive findings within it: schema activation is proven end-to-end, and c33 P1 hydration achieves 100% parameter coverage on both plugins.

Verdict

RENDER_FAILS (VALIDATED at cycle 1; standby-held VALIDATED at cycles 2-3; third consecutive standby resolution).

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/palette_v2_hydration_render_rubric.md	7d8841f089dafd3cfe9ad2bc4e710ddada327c5bf3e3dba9398947b75d7e0
data/palette_v2_render/rubric_hash.txt	7d8841f0...014f
data/palette_v2_render/verdict.json.rubric_hash	7d8841f0...014f

Verified byte-equal in the cycle-1 audit; unchanged across cycles 2-3.

Positive Findings Preserved Within the Negative Finding

The RENDER_FAILS verdict is a fair c31 STILL_GAP extension characterising VST3-binary-internal nondeterminism, not a defect in the c33 P1 hydration workaround or the c34 palette-v2 schema. Substantial positive findings survive:

- **Schema activation end-to-end:** the c34 palette-v2 schema was validated live for the first time on a real render pipeline via `scripts.palette_v2.validate` (READ-ONLY import); assignment builder consumes `format=v2_iterated_params` with c33 Branch B P1-output anchors for Surge XT (bass) and Dexed (other) plus fluidsynth for drums.
- **c33 P1 hydration 100% parameter coverage:** 2855/2855 Surge XT parameters and 2238/2238 Dexed parameters successfully hydrated via `set_parameter(i, v)` iteration from the pinned iterated-params dict. Hydration is not the failure surface.
- **Panel deltas 17-39× the 5% threshold:** `panel_v1_vs_v2` deltas cleared the rubric numeric threshold by a wide margin, confirming that v2 hydration DID move the render audibly relative to v1's fluidsynth fallback. The failure is byte-determinism × 2, not audio motion.

Failure Mechanism (Load-Bearing Finding)

VST3-binary-internal nondeterminism: `run1` and `run2` into fresh `tempfile.mkdtemp()` dirs produce audio that differs at the byte level despite identical hydrated parameter state, identical seed inputs, identical anchored SF2 for drums, and byte-identical assignment JSONL. The failure is inside the Surge XT / Dexed VST3 binaries themselves — a fair extension of the c31 STILL_GAP surface that was originally characterised on `get_state()/save_state()` bindings.

c31 STILL_GAP anti-pattern NOT re-opened: worker did not re-attempt `get_state()`, `save_state(filepath)`, or `set_state(bytes)` (all remain c31 STILL_GAP surface per anti-pattern lock). RENDER_FAILS was emitted honestly with the failure log rather than trying to route around the c31 lock.

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric	Full pipeline; RENDER_FAILS with mechanism honestly diagnosed as VST3-binary-internal nondeterminism; 8 shadow-ledger rows	VALIDATED
2	Verification-only standby	Single-paragraph declaration; canonical rubric SHA cited; zero writes	VALIDATED (standby, no change)
3	Continued standby	One-paragraph declaration; positive findings within negative finding restated; low-output-detector termination	VALIDATED (standby, no change; third consecutive)

State-Machine Discipline (c29 Lemma Respected)

M-DAW-SPIKE-1/palette-schema-v2-hydration-render is a peer sub-milestone under M-DAW-SPIKE-1. NOT a child of terminal-validated M-DAW-SPIKE-1/palette-schema-v2 or its siblings (palette-assignment-schema, palette-instrument-determinism, dawdreamer-state-extraction-workaround).

Ledger Events (Cycle 1: 8 shadow rows under -clone-0 suffix; Cycles 2-3: 0 total)

Six named + two housekeeping, per spec:

1. `_run/cycle_35_launched-clone-0` (status: validated per c35 Branch C newly-codified convention)
2. `_plan/palette_v2_hydration_render_rubric_frozen-clone-0`
3. `_infra/egress-probe-cycle-35-clone-0`
4. M-DAW-SPIKE-1/palette-schema-v2-hydration-render (in-progress; M-* unsuffixed per c32)
5. M-DAW-SPIKE-1/palette-schema-v2-hydration-render (validated verdict roll-up, RENDER_FAILS)
6. `_run/cycle_35_closed-clone-0`
7. `_archive/cycle-35-scratch-clone-0`
8. `_infra/adopt-cycle35-tests-clone-0`

Cycles 2-3: zero across both. validated → in_progress forbidden per c29 lesson.

Anchor Preservation

anchor_preservation.json at cycle 1: unchanged=True across all READ-ONLY anchor families — c34 palette_v2; c33 palette render; c33 dawdreamer state; c31 palette_v1; c31 palette_probe. Preserved unchanged across cycles 2-3 (no workspace state changes). c9 effects chain NOT imported; c13 batch-v2 NOT imported (grep-verified).

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.
- SHA-256 tiebreak; no PRNG (AST-verified); no `sidecar_nonfactor` imports.
- Interpreter guard `assert sys.executable == '/usr/bin/python3'` on every new script.
- Rated audio egress-blocked at *.googlevideo.com (unchanged 403 from c34 baseline). M-EAR-1 armed-not-fired posture holds.
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence:{level,rationale,asses
- UUID5 content-hash `event_id`.
- Merge-report workspace-root fallback carry-forward acknowledged (per c31/c34 pattern).

Anti-Patterns Locked (5-Count Stable; Anti-Pattern #6 Reinforced)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. **c31 STILL_GAP anti-pattern** (`get_state/save_state/set_state(bytes)` re-attempt) explicitly NOT re-opened; `RENDER_FAILS` emitted honestly rather than routing around the lock.

Auditor Guidance (Cycle 3, Verbatim Substance)

Per the researcher's own guidance, residual audit budget should route to sibling clones (clone-1 M-GEN-1/palette-driven-batch-v2-sampler-diversified merge report and clone-2 `_infra/anchor-manifest-v1` merge report both landed) or advance the researcher to cycle 36.

Cycle-36 Handoff (Auditor-Proposed Alternate Line, Reproduced Here)

Emerging c36 candidate line under M-DAW-SPIKE-1 to structurally characterise VST3-binary-internal nondeterminism:

1. **M-DAW-SPIKE-1/dexed-preset-hydration** — probe alternate Dexed preset-load paths (`.syx` `sysex`, JSON preset dump) as candidate replacements for `set_state(bytes)`.
2. **M-DAW-SPIKE-1/surge-xt-fxp-load** — probe Surge XT `.fxp` FX-preset load pathway.
3. **M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization** — bisect the nondeterminism surface (audio callback vs parameter updates vs internal RNG) to localise where the byte drift originates.

These are peer sub-milestones under M-DAW-SPIKE-1, not children of any terminal-validated ancestor; c29 lemma respected.

Preserve locks across c36 descendants: $\alpha = 0.7469387071101908$; NO PRNG; SHA-256 tiebreak only; 5 + 1 anti-patterns locked; `get_state()/save_state(filepath)/set_state(bytes)` remain c31 STILL_GAP surface; `ledger_i3_dminor.jsonl` NOT read.

Merge Disposition

Merge report at `/home/user/music-gen-instance/fork-07063458736e/clone-0/merge_report.md` (spec path; workspace-root fallback also acceptable per c31/c34 pattern). Eight shadow-ledger rows (cycle 1) ready for `concat_clone_ledgers`; zero cross-clone collisions under c32 `-clone-0` suffixes. Cycles 2-3 contribute zero shadow rows across both. Sibling merge reports (clone-1 batch-v2-sampler-diversified; clone-2 anchor-manifest-v1) both landed and ready for root-conductor pickup.

Cumulative Progress

Palette-mechanism scoreboard (post-c35 Branch A): c31 schema validated; c31 instrument determinism validated (sfizz GREEN; Surge XT + Dexed STILL_GAP); c33 clone-1 dawdreamer-state-extraction-workaround WORKAROUND_FOUND (P1 winning); c34 clone-0 palette-schema-v2 SCHEMA_V2_LANDS; c34 clone-1 CROSS_SEED_CONSISTENT; c34 clone-2 BATCH_SPREAD_COLLAPSED; **c35 Branch A palette-schema-v2-hydration-render RENDER_FAILS with VST3-binary-internal nondeterminism identified as fair c31 STILL_GAP extension**; c35 Branch B SPREAD_STILL_COLLAPSED (render_stem API surface); c35 Branch C MANIFEST_LOCKED.

Pattern durability: seven consecutive cycles of rubric-first pre-registration discipline (c26-c30 mechanism probes + c31/c32/c33/c34/c35). Zero rubric-edit-after-analysis incidents. RENDER_FAILS is the first-class negative finding pre-registration is designed to accept — schema activation and 100% hydration coverage survive as positive findings within the negative-finding umbrella. The methodology continues to work exactly as designed.

c29 state-machine lemma respected: every c35 fanout branch is a NEW peer sub-milestone. Ledger topology stays a DAG, not a lineage tree.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard: infra families auto-suffixed -clone-0, substantive M-* unsuffixed — no LedgerConcatError risk at merge.

M-EAR-1 armed-harness Path B: dormant/armed pending audio-egress unblock (still 403 as of c35; retry per policy is non-blocking). **Collision-modeling arc**: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Fanout-harness enhancement candidate (reinforced by this third-consecutive-standby resolution): auto-termination of a clone after N consecutive VALIDATED standby re-invocations (e.g. N = 3) would save ~2-3k tokens per idle cycle. The pattern has now been observed on Branch B c31 (four-cycle chain), c34 clone-0 (four-cycle chain), and c35 clone-0 (this branch, three-cycle chain); the recurrence rate justifies harness codification.

[END OUTPUT]